

ProMeal

Protein-Aware Smart Cooking Assistant

Mobile Computing and Its Applications (2026-1)

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Motivation & Goal

Motivation

- Many people try to meet daily protein intake goals.
- Calculating protein amounts during cooking is inconvenient.
- Users often do not know which ingredients to add when protein is insufficient.

Goal

Develop a mobile app that:

- Detects cooking ingredients from a photo
- Estimates total protein content
- Recommends additional ingredients to meet a target protein intake

Usage Scenario

1. User sets a target protein amount (e.g., 30g)
2. User takes a photo of ingredients
3. The app detects ingredients using YOLOv8
4. Protein content is estimated
5. Additional ingredients are recommended

Key Idea

Single Image → Ingredient Detection → Protein Recommendation Pipeline:

1. Image Capture
2. YOLOv8 Object Detection
3. Ingredient Recognition
4. Protein Lookup Table
5. Protein Deficit Calculation
6. Ingredient Recommendation

The app combines computer vision with nutrition-based recommendation logic.

Dataset

Recipe Ingredients Dataset (Roboflow Universe)

- Total 15,662 images
- 32 ingredient classes
- 640 × 640 images

Dataset Split

- Train: 13,774 images (88%)
- Validation: 794 images (5%)
- Test: 1,094 images (7%)

Preprocessing

- Auto-Orient
- Resize to 640 × 640

Data Augmentation

- Horizontal Flip
- 3 augmented outputs per training image

Comparison of Before / After Model Optimization

Model	Size	mAP@50	mAP@50-95
Before Optimization (FP32)	11.74 MB	93.89%	68.16%
After Optimization (INT8)	3.20 MB	93.48%	67.61%

Quantization Result

- Model size reduced by **72.7%**
- mAP@50 decreased by only **0.41 percentage points**
- INT8 model selected for Android deployment

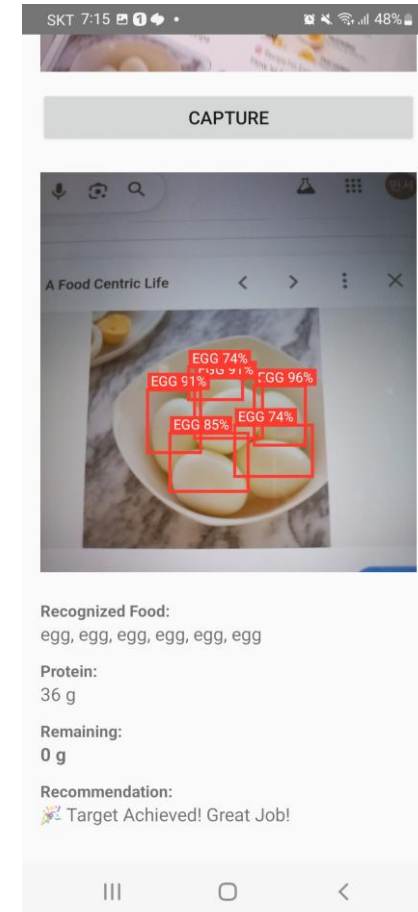
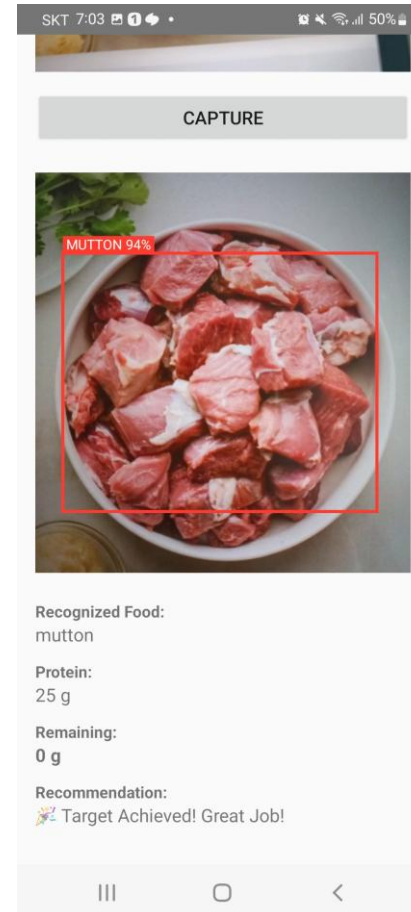
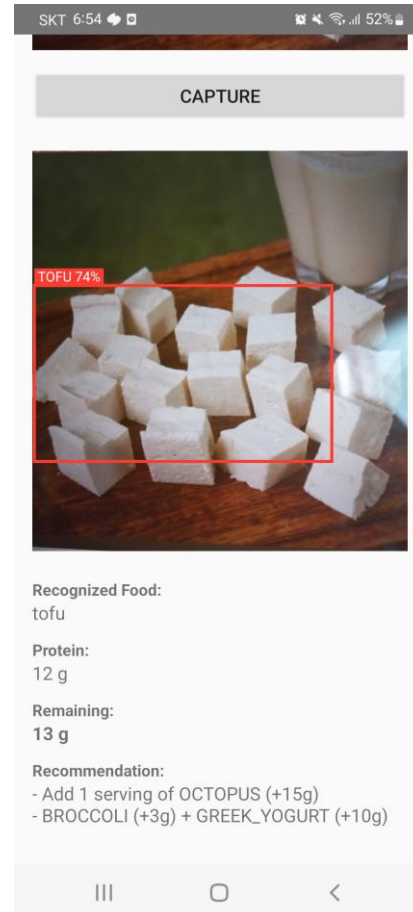
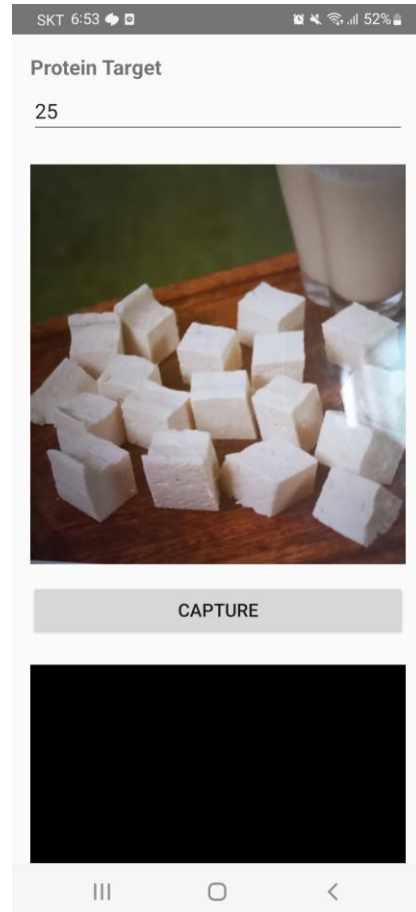
App Functionality Description

Android Application

Features:

- Camera capture
- Ingredient detection
- Bounding box visualization
- Protein estimation
- Ingredient recommendation

Demo



Limitations

- Detection performance decreases when ingredients occupy a small portion of the image.
- Visually similar ingredients (e.g., pork and mutton) may be confused.
- Portion size estimation is not currently supported.

Project Timeline

Week 1

- UI Implementation
- Dataset Selection

Week 2

- YOLOv8 training

Week 3

- TFLite conversion
- INT8 quantization

Week 4

- Android integration
- Testing and evaluation